

three individual sub-pixels—one for each colour—and additional column drivers to select these. A rectangular pixel is split into three sub-pixels, each with a coloured transparent rectangle per sub-pixel.

While the use of TFTs solves many problems, it also creates problems. Major ones are the complex fabrication process and drivers. The high cost of AM TFT LCD panels is a direct result of the complex fabrication processes.

AM TFT panels are superior to passive models in every respect. Flat-panel displays are becoming increasingly commonplace in such devices as cellular phones, personal digital assistants, camcorders and laptops. AM LCDs—ubiquitous in laptops—are the leading flat-panel display technology.

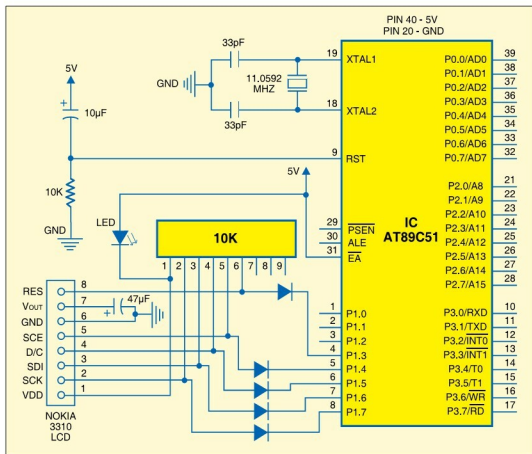


Fig. 1: Schematic for interfacing a Nokia 3310 LCD with AT89C51 microcontroller

Q2. What's the use of microcontrollers in power electronics? How to interface a graphical LCD with microcontrollers?

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A2. Microcontrollers are used to generate timing and control signals in a number of power electronics systems. Some of their applications are:

1. Gate firing control signals in converters
2. Pulse-width modulation
3. Feedback control and processing of feedback signals
4. Implementation of digital controllers and filters
5. Monitoring, sequencing, diagnostics and miscellaneous computation and control
6. Control of AC/DC electric drives
7. Power quality/power factor correction
8. Solar power conditioning

A large number of applications on graphical LCD interfacing with dif-

ferent microcontrollers are available. The schematic for interfacing a Nokia 3310 LCD with AT89C51 microcontroller is shown here in Fig. 1.

Normal LCDs are employed for displaying the characters, whereas graphical LCDs can show images and waveforms in addition to characters.

Q3. (a) In the communication circuit of weaving machines, QTC/Fairchild's 6-pin Optocoupler IC 740L6011 is used. As this product is no longer available, our machines are lying non-operational. Is there any suitable alternative?

(b) For industrial power supply, we are using Texas Instruments' UC2879DW phase-shift-resonant oscillator IC to trigger the SMPS circuit. This part is also unavailable. What would be a suitable replacement?

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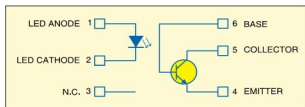


Fig. 2: Pin description of IC 740L6011

A3. (a) IC 740L6011 is a photo-transistor optocoupler/isolator. Its pin diagram along with pin description is shown here in Fig. 2. A number of equivalent ICs with pin compatibility are available in the market—4N25, 4N26, 4N27 and MCT2E just to name a few.

(b) You can try replacing UC2879 with UC1879 or UC3879. Alternatively, if the machine is of high utility, you may consider designing the power supply module with the same functionality and equivalent specifications.

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